

Alex Jacobs

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Education

University of Waterloo

BASc Mechatronics Engineering

Cumulative average 90.79% | cGPA 3.92/4.00 (OMSAS scale)

Sep. 2024 – Expected Apr. 2029

Waterloo, ON

Experience

Mechanical Design Co-op — Quadrupedal & Humanoid Robotics

Sep. 2025 – Dec. 2025

Field AI

Irvine, CA

- Owned mechanical design of a back-mounted sensing and compute payload for the Unitree G1 humanoid from a requirements-only brief, building five 3D-printed iterations in 4 weeks alongside 3 other projects.
- Designed 3 enclosures adding 20 kg of lead shot to a quadruped in 3 days to meet competition weight rules, sized from the $\sim 63.5\%$ random packing fraction; final ballast mass was within 0.2% of target.
- Built custom fixtures to splash-test a sensing payload's enclosures, fans, and speakers against the water half of IP54 (IPX4), identifying insufficient seal clearances that would require a payload redesign.
- Designed and built a modular sensor tower test rig for LiDAR, camera, ultrasonic, and compute payload testing, with adjustable camera mounts and configurable extrinsics in place of the specified fixed design.
- Built a test fixture and Teensy 4.0 data logger to characterize ultrasonic sensor noise across distances, identifying off-axis beam-spread reflections and informing the team's sensor-adoption decision.

Co-Founder & Mechanical Lead — Airborne Object Tracking

Aug. 2025 – Present

Panopsys / Co-op work term, May – Aug. 2026

Toronto, ON

- Co-founded a 3-person venture building a ground-based system that localizes airborne objects in 3D by fusing detections from distributed camera nodes, running on a node-based ROS 2 architecture.
- Own all mechanical design: sensor head, camera and antenna mounts, and compute, RF, and power integration on a portable tripod mast; one latch releases the whole head for transport and setup.
- Led the first outdoor field deployment at an airshow with no structural failures: 3 camera nodes on ~ 100 m baselines captured ~ 3 h of data across 14 display acts, at ranges from overhead out to ~ 5 km.
- Driving the v2 sensor-head design, replacing the 20 mm extrusion frame with a single printed structure to remove the mounting constraints that dictated sensor placement on the v1 prototype.

Research Assistant — Mobile Robotics & SLAM

Jan. 2025 – Apr. 2025

UCLA Sensing and Robotics for Infrastructure Lab / Co-op work term

Los Angeles, CA

- Extended SolidWorks models of an unmanned surface vehicle's above- and below-water sensing kits, including camera mounts, and designed a 3D-printed hydrodynamic fairing for its underwater sensors.
- Benchmarked 5+ open-source SLAM systems, including DLIO, LIO-SAM, Coco-LIC, and ORB-SLAM, on datasets outside their design domains, characterizing localization and mapping failure modes.
- Built a Gazebo simulation for a quadruped from existing campus models via a Blender conversion pipeline, and wrote Python to add Gaussian noise to its otherwise ideal simulated sensor readings.

Technical Skills

CAD & Modeling: SolidWorks, Onshape, Fusion 360, Blender

Mechanical Design: enclosures, mounts and fixtures, cable routing, design for 3D printing

Fabrication & Test: FDM 3D printing; splash testing, sensor characterization, field testing

Software & Electronics: ROS 2, Gazebo, Python, C++, SLAM evaluation; Teensy, wiring harnesses

Airborne Object Tracking System

Panopsys / Co-Founder & Mechanical Lead / Aug. 2025 – Present



Node tracking a CP-140 Aurora at the airshow



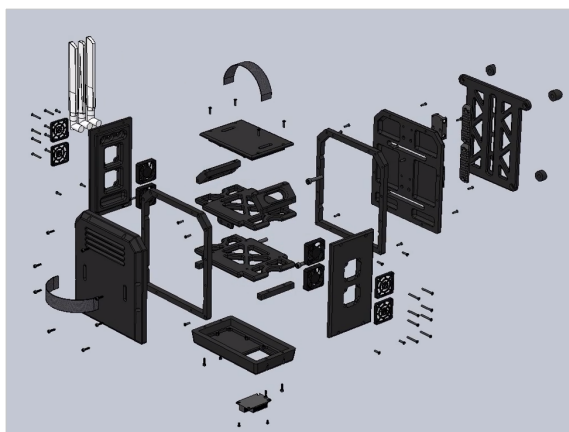
Camera module and radio hardware on the mast

First field deployment: 3 camera nodes | ~100 m baselines | 14 display acts | out to ~5 km | ~3 h capture

- *Goal:* Locate airborne objects in 3D by fusing detections from distributed ground camera nodes on ROS 2.
- *Design:* As co-founder, own all mechanical design: camera housings and mounts, antenna placement, sensor-head structure, and integration of compute, RF, and power hardware on a portable tripod mast.
- *Design:* One latch frees the whole 20 mm extrusion head from the tripod for fast transport and setup.
- *Result:* Led the first outdoor deployment at an airshow, with no structural failures over ~3 h of capture. It surfaced issues to fix before the next deployment, including an unseated antenna that blocked an RTK fix.
- *Next:* The v2 sensor head, now in progress, replaces the 20 mm extrusion frame with a single printed structure, removing the mounting constraints that dictated sensor placement on the v1 prototype.

Back-Mounted Sensing and Compute Payload

Field AI / Mechanical Design Co-op / Nov. 2025 – Dec. 2025



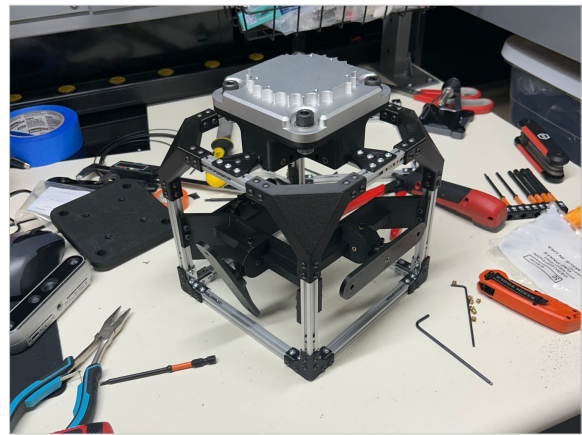
Exploded view of my payload design



Fabricated prototypes for cross-office testing

- *Goal:* Port existing sensing and compute hardware onto the Unitree G1 humanoid, from a requirements-only brief.
- *Design:* Architected the frame, modular shelf layout, and mounting interface, with internal cable routing and thermal integration so the hardware dropped in cleanly and its connectors and fans stayed serviceable.
- *Result:* Built five 3D-printed iterations in 4 weeks, alongside 3 other projects, for testing at the Boston office, and wrote the design documentation and file-structure guides for handoff to downstream engineers.

- *Goal:* Build a modular tower for testing LiDAR, camera, ultrasonic, and compute payload setups.
- *Design:* Chose adjustable camera mounts with configurable extrinsics over the specified fixed design.
- *Design:* Built a fixture holding 2 perpendicular ultrasonic sensors at fixed, known offsets, so distance was the only thing that changed between runs, plus a Teensy 4.0 logger recording their readings over time.
- *Result:* Found off-axis beam-spread reflections in the noise, informing the team's adoption decision.



The sensor tower test rig I designed and built

Competition Ballast & Ingress Protection

Field AI / Mechanical Design Co-op / Sep. 2025 – Dec. 2025

Competition ballast

- *Goal:* Add 20 kg to a quadruped in 3 days to meet weight rules at an international event in Singapore.
- *Design:* Sized 3 lead-shot enclosures for minimum volume, $V = m/(\phi\rho_{\text{Pb}})$, at random packing $\phi \approx 0.635$.
- *Result:* Final ballast mass within 0.2% of target.

IP54 feasibility

- *Goal:* Check if a sensing payload could reach IP54.
- *Design:* Built splash-test fixtures for its enclosures, fans, and speakers, covering IP54's water half (IPX4).
- *Result:* Found insufficient seal clearances, meaning the payload would need a redesign to reach IP54.

Unmanned Surface Vehicle

UCLA Sensing and Robotics for Infrastructure Lab / Aug. 2023, Jan. – Apr. 2025



The lab's USV; my work was its sensing-kit models and fairing

- *Goal:* Support the lab's evolving sensor configurations on its unmanned surface vehicle (USV).
- *Design:* Modeled the hull on a 2023 high-school placement, then returned in 2025 to extend SolidWorks models of the above- and below-water sensing kits, including hull parts and camera mounts.
- *Design:* Modeled and 3D-printed a hydrodynamic fairing to reduce drag on the underwater sensors.

Simulation and SLAM Evaluation

UCLA Sensing and Robotics for Infrastructure Lab / Jan. – Apr. 2025

- *Design:* Built a Gazebo simulation for a quadruped from campus models via Blender, with Python adding Gaussian noise to otherwise ideal sensor readings.
- *Design:* Deployed SLAM packages on lab workstations, patching their source code and build dependencies to resolve compatibility issues.
- *Result:* Benchmarked 5+ open-source SLAM systems, including DLIO, LIO-SAM, Coco-LIC, and ORB-SLAM, on out-of-domain datasets, comparing point clouds and localization to document failures.